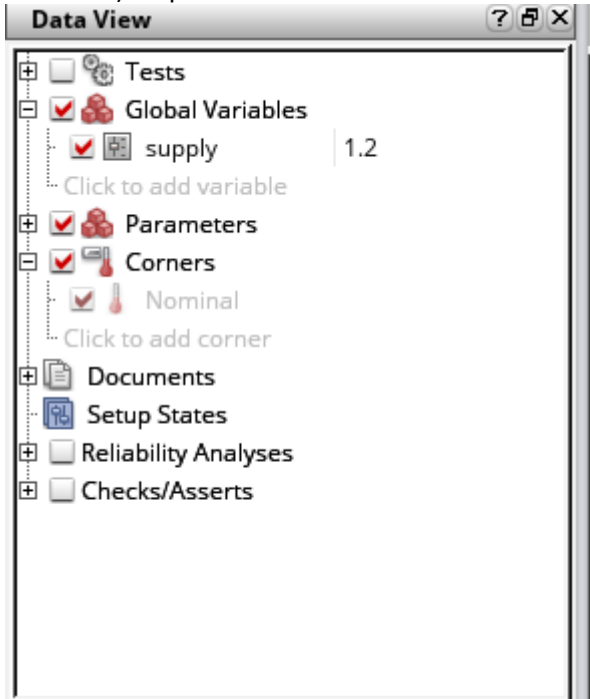


Lab 09**Feedback and PVT & MC sims****PART 1: Feedback**

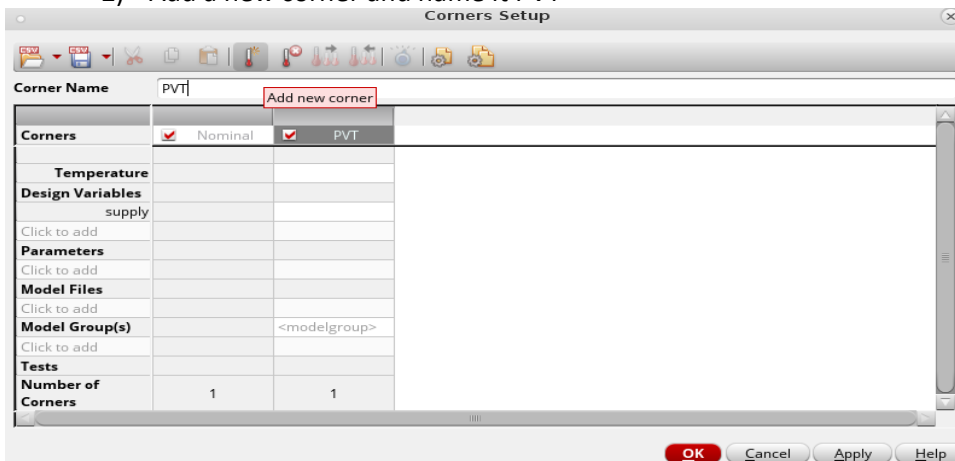
- 1) We'll use the schematic of the 5T-OTA designed in lab 05 to study feedback
- 2) Connect the output node to one of the input nodes to create unity gain feedback (which input should you use to connect to the output to get negative feedback?)
- 3) Report closed loop gain vs frequency. (Hint: Run AC simulation 1Hz:10Gz, logarithmic, 10 points/decade). Indicate the DC gain and the bandwidth.
- 4) Report loop gain vs frequency, In our case the loop gain is the open loop DC gain of the OTA. Annotate the DC gain, the dominant pole, and the unity gain frequency in the plot.
- 5) Change the input source to sine wave of frequency $F_{in}=1\text{kHz}$ and magnitude 50 mV and run transient analysis for $\frac{5}{F_{in}}$ seconds (in case of 1kHz it'll be 5 mS). Plot the input signal, the output signal, and the differential input signal of the OTA ($V_P - V_N$), Calculate the peak-to-peak voltage of the previous three signals. What is the relation between the output and ($V_P - V_N$)? Comment.
- 6) Change the frequency $F_{in}=5\text{MHz}$ and magnitude 50 mV and run transient analysis for $\frac{5}{F_{in}}$ seconds (in case of 5MHz it'll be 1 μS). Plot the input signal, the output signal, and the differential input signal of the OTA ($V_P - V_N$), Calculate the peak-to-peak voltage of the previous three signals. What is the relation between the output and ($V_P - V_N$)? Comment and compare to the 1kHz case.

Part 2: PVT & MC sims

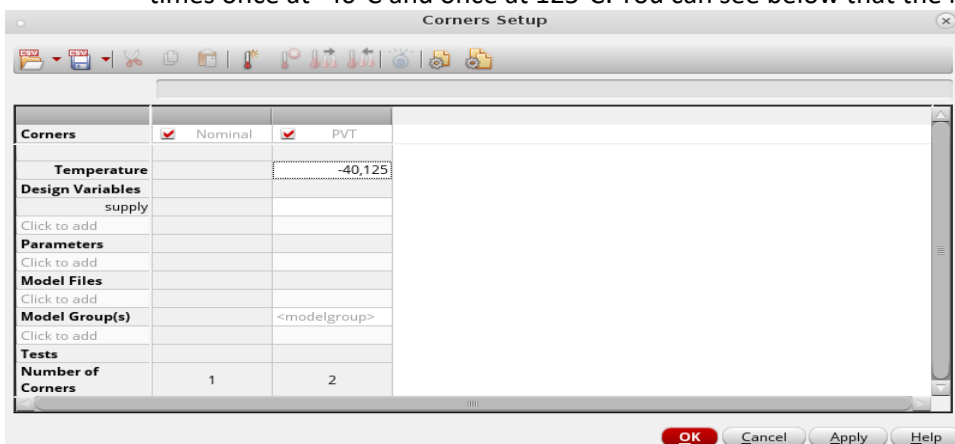
- 1) Open ADEXL view and select “click to add corner”



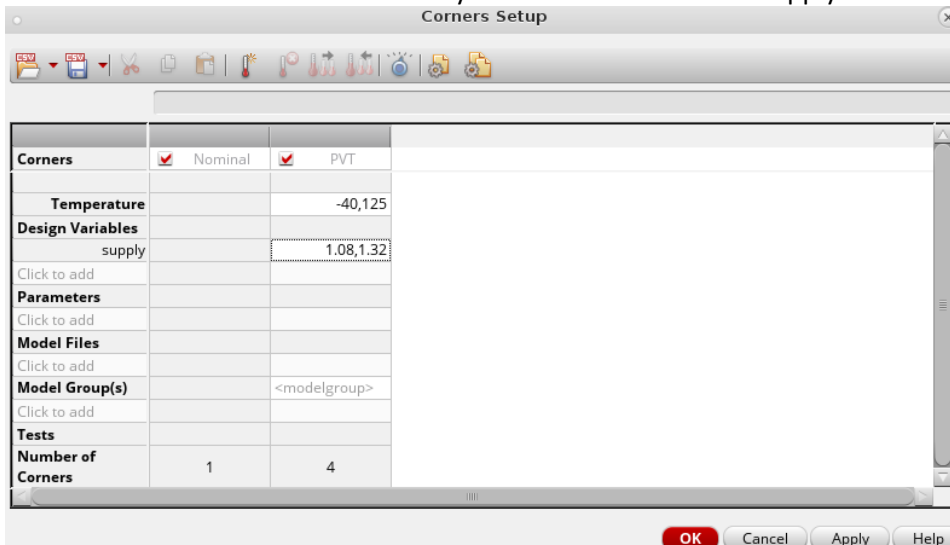
- 2) Add a new corner and name it PVT



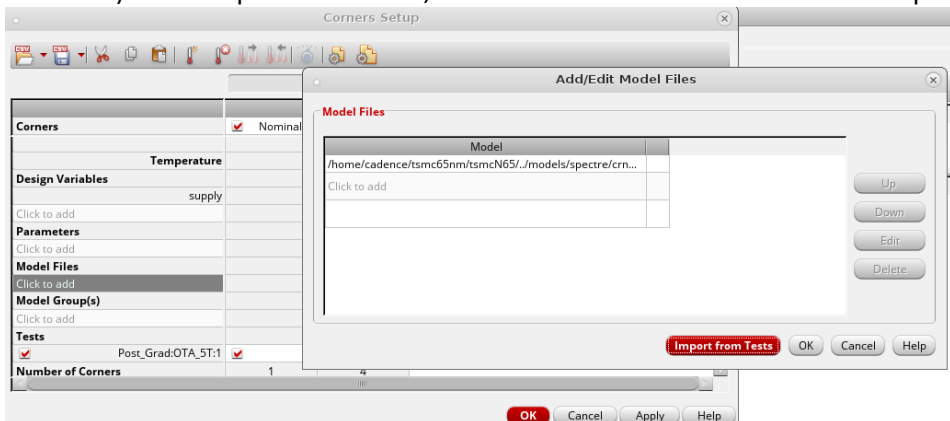
- 3) In Temperature add the temperature range (from -40°C to 125°C). This will run the design two times once at -40°C and once at 125°C. You can see below that the number of corners is 2



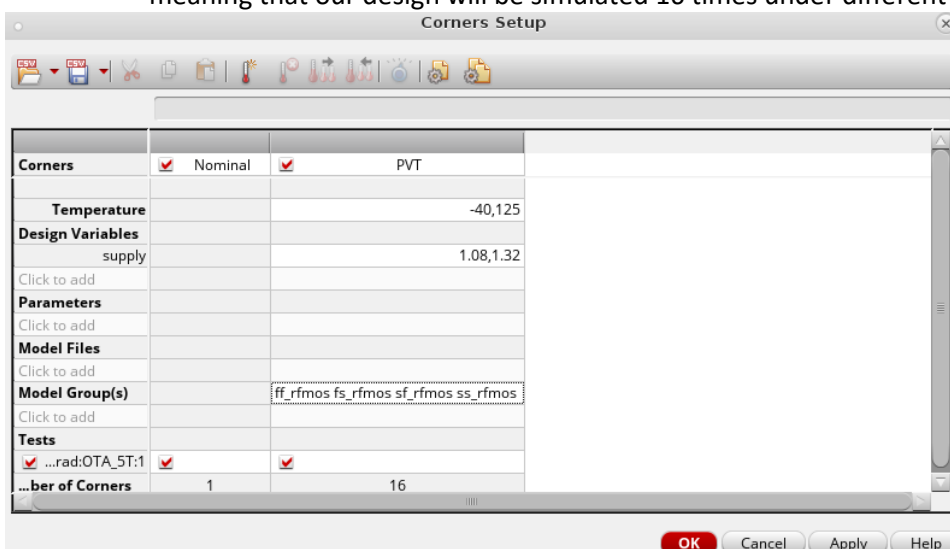
- 4) Click on “click to add” in Design Variables and add the variable that represents your supply and add the supply range (for example from 1.08 to 1.32). This will run the design two times once at supply equal 1.08 V and once at supply equal 1.32V. You can see below that the number of corners is now 4 because you have two corners for supply and two corners for temperature



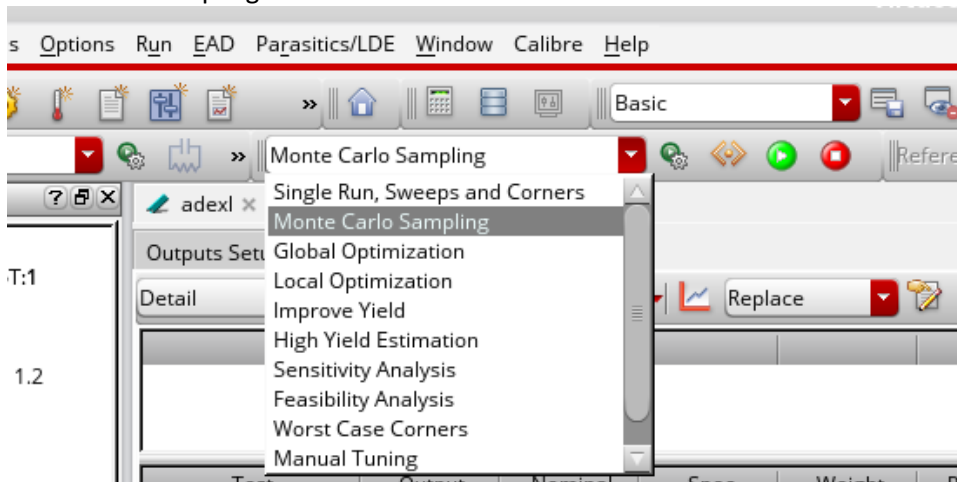
- 5) To add process corners, click to add model files then choose “import from Tests” then press ok



- 6) Then add ff, fs, sf and ss corners as shown below. Notice that number of corners now is 16, meaning that our design will be simulated 16 times under different PVT conditions



- 7) To run MC, First disable corners and any sweeps then change the models to MC models that contains mismatch info then change “Single Run, Sweeps and Corners” to “Monte Carlo Sampling”



- 8) Then click on “simulation options” and adjust the setting as shown below

